



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**PHYSICAL SCIENCES: PHYSICS (P1)
FISIESE WETENSKAPPE: FISIKA (V1)**

2023

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 30 pages./
Hierdie nasienriglyne bestaan uit 30 bladsye.**

GENERAL MARKING GUIDELINES PAPER 1 ALGEMENE NASIEN RIGLYNE VRAESTEL 1

1. CALCULATIONS/BEREKENINGE

- 1.1 **Marks will be awarded for:** correct formula, correct substitution, correct answer with unit.
Punte sal toegeken word vir: korrekte formule, korrekte substitusie, korrekte antwoord met eenheid.
- 1.2 **No marks will be awarded if an incorrect or inappropriate formula is used**, even though there may be relevant symbols and applicable substitutions.
Geen punte sal toegeken word waar 'n verkeerde of ontoepaslike formule gebruik word nie, selfs al is daar relevante simbole en relevante substitusies.
- 1.3 When an error is made during **substitution into a correct formula**, a mark will be awarded for the correct formula and for the correct substitutions, but **no further marks** will be given.
*Wanneer 'n fout gedurende **substitusie in 'n korrekte formule** begaan word, sal 'n punt vir die korrekte formule en vir die korrekte substitusies toegeken word, maar **geen verdere punte** sal toegeken word nie.*
- 1.4 If **no formula** is given, but all **substitutions are correct**, a candidate will **forfeit one mark**.
*Indien **geen formule** gegee is nie, maar al die substitusies is korrek, verloor die kandidaat een punt.*
- 1.5 **No penalisation if zero substitutions are omitted** in calculations where **correct formula/principle is given correctly**.
Geen penalisering indien nulwaardes nie getoon word nie in berekeninge waar die *formule/beginsel* korrek gegee is nie
- 1.6 Mathematical manipulations and change of subject of appropriate formulae carry no marks, but if a candidate starts off with the correct formula and then changes the subject of the formula incorrectly, marks will be awarded for the formula and the correct substitutions. The mark for the incorrect numerical answer is forfeited.
Wiskundige manipulasies en verandering van onderwerp van toepaslike formules tel geen punte nie, maar indien 'n kandidaat met die korrekte formule begin en dan die onderwerp van die formule verkeerd verander, sal punte vir die formule en korrekte substitusies toegeken word. Die punt vir die verkeerde numeriese antwoord word verbeur

- 1.7 Marks are only awarded for a formula if a **calculation has been attempted**, i.e. substitutions have been made or a numerical answer given.
*Punte word slegs vir 'n formule toegeken indien 'n **poging tot 'n berekening aangewend is**, d.w.s. substitusies is gedoen of 'n numeriese antwoord is gegee.*
- 1.8 Marks can only be allocated for substitutions when values are substituted into formulae and not when listed before a calculation starts.
Punte kan slegs toegeken word vir substitusies wanneer waardes in formules ingestel is en nie vir waardes wat voor 'n berekening gelys is nie.
- 1.9 All calculations, when not specified in the question, must be rounded off to a minimum of TWO decimal places.
Alle berekenings, wanneer dit nie in die vraag gespesifiseer word nie, moet tot 'n minimum van TWEE desimale plekke afgerond word.
- 1.10 If a final answer to a calculation is correct, full marks will not automatically be awarded. Markers will always ensure that the correct/appropriate formula is used and that workings, including substitutions, are correct.
Indien 'n finale antwoord van 'n berekening korrek is, sal volpunte nie outomaties toegeken word nie. Nasieners sal altyd verseker dat die korrekte/toepaslike formule gebruik word en dat bewerkings, insluitende substitusies, korrek is
- 1.11 Questions where a series of calculations do not necessarily always have to follow the same order (as in circuit calculations), full marks will be awarded provided that it is a valid solution to the problem.
Vrae waar 'n reeks berekeninge nie noodwendig altyd in dieselfde volgorde hoef te wees nie (soos in stroombaanberekeninge) sal volpunte toegeken word op voorwaarde dat dit 'n geldige oplossing vir die probleem is.
- 1.12 Any calculation that will not bring the candidate closer to the answer than the original solution, will not count any marks.
Enige berekening wat nie die kandidaat nader aan die antwoord as die oorspronklike oplossing bring nie, sal geen punte tel nie.

2. **Units/Eenhede**

- 2.1 Candidates will only be penalised once for the repeated use of an incorrect unit **within a question**.
*Kandidate sal slegs een keer gepeenaliseer word vir die herhaaldelike gebruik van 'n verkeerde **eenheid in 'n vraag**.*
- 2.2 Units are only required in the final answer to a calculation.
Eenhede word slegs in die finale antwoord op 'n berekening verlang.

- 2.3 Marks are only awarded for an answer, and not for a unit per se. Candidates will therefore forfeit the mark allocated for the answer in each of the following situations:

- Correct answer + wrong unit
- Wrong answer + correct unit
- Correct answer + no unit

Punte word slegs vir 'n antwoord en nie vir 'n eenheid op sigself toegeken nie. Kandidate sal dus die punt wat toegeken is vir die antwoord in elk van die volgende gevalle verbeur:

- Korrekte antwoord + verkeerde eenheid
- Verkeerde antwoord + korrekte eenheid
- Korrekte antwoord + geen eenheid

- 2.4 SI units must be used, except in certain cases, e.g. $V \cdot m^{-1}$ instead of $N \cdot C^{-1}$, and $cm \cdot s^{-1}$ or $km \cdot h^{-1}$ instead of $m \cdot s^{-1}$ where the question warrants this. *SI-eenhede moet gebruik word, behalwe in sekere gevalle, bv. $V \cdot m^{-1}$ in plaas van $N \cdot C^{-1}$, en $cm \cdot s^{-1}$ of $km \cdot h^{-1}$ in plaas van $m \cdot s^{-1}$ waar die vraag dit regverdig.*

3

General/Algemeen

- 3.1 If one answer or calculation is required, but two are given by the candidate, only the first one will be marked, irrespective of which one is correct. If two answers are required, only the first two will be marked, etc.

Indien een antwoord of berekening verlang word, maar twee word deur die kandidaat gegee, sal slegs die eerste een nagesien word, ongeag watter een korrek is. Indien twee antwoorde verlang word, sal slegs die eerste twee nagesien word, ens.

- 3.2 For marking purposes, alternative symbols (s, u, t, etc.) will also be accepted.

Vir nasiendoeleindes sal alternatiewe simbole (s, u, t, ens.) ook aanvaar word.

- 3.3 Separate compound units with a multiplication dot, not a full stop, e.g. $m \cdot s^{-1}$. For marking purposes, $m \cdot s^{-1}$ and m/s will also be accepted.

Skei saamgestelde eenhede met 'n vermenigvuldigpunt, nie met 'n punt nie, bv. $m \cdot s^{-1}$. Vir nasiendoeleindes sal $m \cdot s^{-1}$ en m/s ook aanvaar word.

4. Positive marking/Positiewe nasien

Positive marking regarding calculations will be followed in the following cases:
Positiewe nasien met betrekking tot berekenings sal in die volgende gevalle geld:

- 4.1 **Subquestion to subquestion:** When a certain variable is incorrectly calculated in one subquestion (e.g. 3.1) and needs to be substituted into another subquestion (3.2 or 3.3), **full marks** are to be awarded for the subsequent subquestions.

Subvraag na subvraag: Wanneer 'n sekere veranderlike in een subvraag (bv. 3.1) bereken word en dan in 'n ander vervang moet word (3.2 of 3.3), word **volpunte** vir die daaropvolgende subvrae toegeken.

- 4.2 **A multistep question in a subquestion:** If the candidate has to calculate, for example, current in the first step and gets it wrong due to a substitution error, the mark for the substitution and the final answer will be forfeited.
'n Vraag met veelvuldige stappe in 'n subvraag: Indien 'n kandidaat, byvoorbeeld, die stroom verkeerd bereken in die eerste stap as gevolg van 'n substitusiefout, verbeur die kandidaat die punt vir die substitusie sowel as die finale antwoord.

5. Negative marking/Negatiewe nasien

Normally an incorrect answer cannot be correctly motivated if based on a conceptual mistake. If the candidate is therefore required to motivate in QUESTION 3.2 the answer given to QUESTION 3.1, and QUESTION 3.1 is incorrect, no marks can be awarded for QUESTION 3.2. However, if the answer for, for example, QUESTION 3.1 is based on a calculation, the motivation for the incorrect answer in QUESTION 3.2 should be considered.
'n Verkeerde antwoord, indien dit op 'n konsepsuele fout gebaseer is, kan nie korrek gemotiveer word nie. Indien die kandidaat dus gevra word om in VRAAG 3.2 die antwoord op VRAAG 3.1 te motiveer en VRAAG 3.1 is verkeerd, kan geen punte vir VRAAG 3.2 toegeken word nie. Indien die antwoord op, byvoorbeeld, VRAAG 3.1 egter op 'n berekening gebaseer is, moet die motivering vir die verkeerde antwoord in VRAAG 3.2 oorweeg word.

QUESTION 1/VRAAG 1

- | | | |
|------|---------------------------|-------------|
| 1.1 | A ✓✓ | (2) |
| 1.2 | B ✓✓ (accept Q/aanvaar Q) | (2) |
| 1.3 | D ✓✓ | (2) |
| 1.4 | D ✓✓ | (2) |
| 1.5 | D ✓✓ | (2) |
| 1.6 | B OR/OF D ✓✓ | (2) |
| 1.7 | B ✓✓ | (2) |
| 1.8 | A ✓✓ | (2) |
| 1.9 | A ✓✓ | (2) |
| 1.10 | D ✓✓ | (2) |
| | | [20] |

QUESTION 2/VRAAG 2

2.1

Marking criteria/Nasienkriteria - Formula to calculate a./Formule om a te bereken. ✓ - Correct substitution to calculate a./Korrekte vervanging om a te bereken. ✓✓	
OPTION 1/OPSIE 1 DOWNWARDS AS POSITIVE/ AFWAARTS AS POSITIEF $v_f^2 = v_i^2 + 2a\Delta y$ ✓ $(3,41)^2 \checkmark = (0)^2 + (2)a(1,5)$ ✓ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$	UPWARDS AS POSITIVE/ OPWAARTS AS POSITIEF $v_f^2 = v_i^2 + 2a\Delta y$ ✓ $(-3,41)^2 \checkmark = (0)^2 + (2)a(-1,5)$ ✓ $a = -3,88$ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$
OPTION 2/OPSIE 2 DOWNWARDS AS POSITIVE/ AFWAARTS AS POSITIEF $\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t$ $1,5 = \left(\frac{0 + 3,41}{2} \right) \Delta t$ $\Delta t = 0,88 \text{ s}$ UPWARDS AS POSITIVE/ OPWAARTS AS POSITIEF $\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t$ $-1,5 = \left(\frac{0 - 3,41}{2} \right) \Delta t$ $\Delta t = 0,88 \text{ s}$	$v_f = v_i + a\Delta t$ ✓ $-3,41 \checkmark = (0) + a(0,88)$ ✓ $a = -3,88$ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$ OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓ $-1,5 \checkmark = (0)(0,88) + \frac{1}{2}a(0,88)^2$ ✓ $a = -3,88$ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$ $v_f = v_i + a\Delta t$ ✓ $3,41 \checkmark = (0) + a(0,88)$ ✓ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$ OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓ $1,5 \checkmark = (0)(0,88) + \frac{1}{2}a(0,88)^2$ ✓ $a = 3,88 \text{ m}\cdot\text{s}^{-2}$

(3)

2.2

**ACCEPT/AANVAAR**

	Accepted symbols/Aanvaarde simbole
W ✓	F_g/F_w /weight/gewig/mg/gravitational force/gravitasiekrag/ $F_{\text{Earth on block}}/F_{\text{Aarde op blok}}/73,5N$
T ✓	Tension/Spinning/ $F_{\text{Tension}}/F_{\text{Spanning}}/F_{\text{rope}}/F_{\text{tou}}/F_T/F$
Notes/Aantekeninge: - Mark awarded for label <u>and</u> arrow./Punt toegeken vir byskrif <u>en</u> pyltjie. - Do not penalise for length of arrows since drawing is not to scale./Moenie vir die lengte van die pyltjies penaliseer nie aangesien die tekening nie volgens skaal is nie. - Any other additional force(s)/Enige ander addisionele krag(te): Max/Maks $\frac{1}{2}$	

(2)

2.3

Marking criteria/Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark./Indien enige van die onderstreepte sleutelwoorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

When a resultant/net force acts on an object, the object will accelerate in the direction of the force with an acceleration that is directly proportional to the force and inversely proportional to the mass of the object. ✓✓

Wanneer 'n resulterende/netto krag op 'n voorwerp inwerk, sal die voorwerp in die rigting van die krag versnel teen 'n versnelling wat direk eweredig aan die krag en omgekeerd eweredig aan die massa van die voorwerp is.

OR/OF

The resultant/net force acting on an object is equal to the rate of change of momentum of the object in the direction of the resultant/net force.

Die resulterende/netto krag wat op 'n voorwerp inwerk is gelyk aan die tempo van verandering van momentum van die voorwerp in dieselfde rigting as die resulterende/netto krag.

(2)

2.4

Marking criteria/Nasienkriteria

- Any correct formula./Enige korrekte formule. ✓
- Correct substitution to calculate tension./Korrekte vervanging om spanning te bereken. ✓✓
- Correct substitution to calculate mass of block A./Korrekte vervanging om massa van blok A te bereken. ✓
- Correct final answer/Korrekte finale antwoord: 3,25 kg ✓

Calculation of tension (Block B) 3 marks:
Berekening van spanning (Blok B) 3 punte:
DOWNWARDS POSITIVE/AFWAARTS POSITIEF

$$\left. \begin{array}{l} F_{\text{net}} = ma \\ F_g + T = ma \\ mg - T = ma \end{array} \right\} \begin{array}{l} \checkmark \text{Any one/} \\ \text{Enige een} \end{array}$$

$$\underline{7,5(9,8) - T} \checkmark = \underline{7,5(3,88)} \checkmark$$

$$T = 44,40 \text{ N}$$

UPWARDS POSITIVE/OPWAARTS POSITIEF

$$\left. \begin{array}{l} F_{\text{net}} = ma \\ T - F_g = ma \\ T - mg = ma \end{array} \right\} \begin{array}{l} \checkmark \text{Any one/} \\ \text{Enige een} \end{array}$$

$$\underline{T - 7,5(9,8)} \checkmark = \underline{7,5(-3,88)} \checkmark$$

$$T = 44,40 \text{ N}$$

Calculation of mass**(Block A) 2 marks:**
Berekening van massa
(Blok A) 2 punte:
UPWARDS POSITIVE
OPWAARTS POSITIEF

$$\begin{array}{l} F_{\text{net}} = ma \\ T - F_g = ma \\ T - mg = ma \end{array}$$

$$\underline{44,40 - m(9,8) = m(3,88)} \checkmark$$

$$m = 3,25 \text{ kg} \checkmark$$

DOWN POSITIVE
AF POSITIEF

$$\begin{array}{l} F_{\text{net}} = ma \\ F_g - T = ma \\ mg - T = ma \end{array}$$

$$\underline{m(9,8) - 44,40 = m(-3,88)} \checkmark$$

$$m = 3,25 \text{ kg} \checkmark$$

(5)

2.5

Marking criteria/Nasienkriteria

- Any correct formula./Enige korrekte formule. ✓
- Correct substitution of v_i and v_f ./Korrekte vervanging van v_i en v_f . ✓
- Correct substitution of $9,8 \text{ m} \cdot \text{s}^{-2}$./Korrekte vervanging van $9,8 \text{ m} \cdot \text{s}^{-2}$. ✓
- Adding 1,5 m to calculated Δy . /Tel 1,5 m by berekende Δy . ✓
- Correct final answer/Korrekte finale antwoord: 2,09 m ✓

OPTION 1/OPSIE 1**UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF**

$$v_f^2 = v_i^2 + 2a\Delta y \quad \checkmark$$

$$(0^2) = (3,41)^2 \checkmark + (2)(-9,8)\Delta y \quad \checkmark$$

$$\Delta y = 0,59 \text{ m}$$

$$\text{Maximum height} = 0,59 + 1,5 \quad \checkmark$$

$$= 2,09 \text{ m} \quad \checkmark$$

DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF

$$v_f^2 = v_i^2 + 2a\Delta y \quad \checkmark$$

$$(0^2) = (-3,41)^2 \checkmark + (2)(9,8)\Delta y \quad \checkmark$$

$$\Delta y = -0,59$$

$$\Delta y = 0,59 \text{ m}$$

$$\text{Maximum height} = 0,59 + 1,5 \quad \checkmark$$

$$= 2,09 \text{ m} \quad \checkmark$$

OPTION 2/OPSIE 2**UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF**

$$v_f = v_i + a\Delta t$$

$$0 = 3,41 + (-9,8)\Delta t$$

$$\Delta t = 0,35 \text{ s}$$

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \quad \checkmark$$

$$= (3,41)(0,35) \checkmark + \frac{1}{2}(-9,8)(0,35)^2 \quad \checkmark$$

$$= 0,59 \text{ m}$$

$$\text{Maximum height} = 0,59 + 1,5 \quad \checkmark$$

$$= 2,09 \text{ m} \quad \checkmark$$

DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF

$$v_f = v_i + a\Delta t$$

$$0 = -3,41 + (9,8)\Delta t$$

$$\Delta t = 0,35 \text{ s}$$

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \quad \checkmark$$

$$= (-3,41)(0,35) \checkmark + \frac{1}{2}(9,8)(0,35)^2 \quad \checkmark$$

$$= -0,59$$

$$\Delta y = 0,59 \text{ m}$$

$$\text{Maximum height} = 0,59 + 1,5 \quad \checkmark$$

$$= 2,09 \text{ m} \quad \checkmark$$

OPTION 3/OPSIE 3 UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF $v_f = v_i + a\Delta t$ $0 = 3,41 + (-9,8)\Delta t$ $\Delta t = 0,35 \text{ s}$ $\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t \checkmark$ $\Delta y = \left(\frac{3,41 + 0}{2} \right) \checkmark (0,35) \checkmark$ $= 0,59 \text{ m}$ Maximum height = $0,59 + 1,5 \checkmark$ $= 2,09 \text{ m} \checkmark$	DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF $v_f = v_i + a\Delta t$ $0 = -3,41 + (9,8)\Delta t$ $\Delta t = 0,35 \text{ s}$ $\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t \checkmark$ $\Delta y = \left(\frac{-3,41 + 0}{2} \right) \checkmark (0,35) \checkmark$ $= -0,59 \text{ m}$ Maximum height = $0,59 + 1,5 \checkmark$ $= 2,09 \text{ m} \checkmark$
Note/Aantekening: OPTION 4 TO 5/OPSIE 4 TOT 5 Substitution of incorrect mass/Vervanging van verkeerde massa: $max/maks: \frac{3}{5}$	
OPTION 4/OPSIE 4 $\left. \begin{aligned} (E_{\text{mech}})_{\text{top}} &= (E_{\text{mech}})_{\text{bottom}} \\ (E_p + E_k)_{\text{top}} &= (E_p + E_k)_{\text{bottom}} \\ (mgh + \frac{1}{2}mv_i^2)_{\text{top}} &= (mgh + \frac{1}{2}mv_f^2)_{\text{bottom}} \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$ $(9,8)(h) + (0) \checkmark = (0) + \frac{1}{2}(3,41)^2 \checkmark$ $h = 0,59 \text{ m}$ Maximum height = $0,59 + 1,5 \checkmark$ $= 2,09 \text{ m} \checkmark$	
OPTION 5/OPSIE 5 $\left. \begin{aligned} W_{\text{nc}} &= \Delta K + \Delta U \\ W_{\text{nc}} &= \Delta K + mg(h_f - h_i) \\ 0 &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 + mgh_f - mgh_i \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$ $(0) = (0) - \frac{1}{2}(3,41)^2 \checkmark + (9,8)(h) \checkmark$ $h = 0,59 \text{ m}$ Maximum height = $0,59 + 1,5 \checkmark$ $= 2,09 \text{ m} \checkmark$	
OPTION 6/OPSIE 6 $\left. \begin{aligned} W_{\text{net}} &= \Delta E_k \\ w\Delta y \cos\theta &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$ $(9,8)(\Delta y) \cos 180^\circ \checkmark = 0 - \frac{1}{2}(3,41)^2 \checkmark$ $\Delta y = 0,59 \text{ m}$ Maximum height = $0,59 + 1,5 \checkmark$ $= 2,09 \text{ m} \checkmark$	

(5)
[17]

QUESTION 3/VRAAG 3

- 3.1 Motion under the influence of gravity/weight/gravitational force only. ✓✓
 Beweging slegs onder die invloed van gravitasie/gewig/gravitasiekrag.

(2 or/of 0)**OR/OF**Motion in which the only force acting is the gravitational force.Beweging waar die enigste krag wat inwerk gravitasiekrag is.**(2)**

3.2.1

Marking criteria/Nasienkriteria

- Formula to calculate Δt /Formule om Δt te bereken. ✓
- Correct substitution to calculate Δt /Korrekte vervanging om Δt te bereken. ✓
- Final answer/Finale antwoord: 1,76 s ✓

OPTION 1/OPSIE 1**UPWARDS AS POSITIVE/
OPWAARTS AS POSITIEF**

$$\Delta y = v_i \Delta t + \frac{1}{2} a \Delta t^2 \checkmark$$

$$-15,2 = (0) + \frac{1}{2}(-9,8)\Delta t^2 \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

**DOWNWARDS AS POSITIVE/
AFWAARTS AS POSITIEF**

$$\Delta y = v_i \Delta t + \frac{1}{2} a \Delta t^2 \checkmark$$

$$15,2 = (0) + \frac{1}{2}(9,8)\Delta t^2 \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

OPTION 2/OPSIE 2**UPWARDS AS POSITIVE/
OPWAARTS AS POSITIEF**

$$v_f^2 = v_i^2 + 2a\Delta y$$

$$v_f^2 = (0)^2 + (2)(-9,8)(-15,2)$$

$$v_f = -17,26 \text{ m} \cdot \text{s}^{-1} \longrightarrow$$

**UPWARDS AS POSITIVE/
OPWAARTS AS POSITIEF**

$$v_f = v_i + a\Delta t \checkmark$$

$$-17,26 = (0) + (-9,8)\Delta t \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

**DOWNWARDS AS POSITIVE/
AFWAARTS AS POSITIEF**

$$v_f^2 = v_i^2 + 2a\Delta y$$

$$v_f^2 = (0)^2 + (2)(9,8)(15,2)$$

$$v_f = 17,26 \text{ m} \cdot \text{s}^{-1} \longrightarrow$$

OR/OF

$$\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t \checkmark$$

$$-15,2 = \left(\frac{0 - 17,26}{2} \right) \Delta t \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

OPTION 3/OPSIE 3

$$(E_{\text{mech}})_{\text{top}} = (E_{\text{mech}})_{\text{bottom}}$$

$$(E_p + E_k)_{\text{top}} = (E_p + E_k)_{\text{bottom}}$$

$$(mgh + \frac{1}{2}mv_i^2)_{\text{top}} = (mgh + \frac{1}{2}mv_f^2)_{\text{bottom}}$$

$$(9,8)(15,2) + 0 = 0 + (\frac{1}{2})(v_f)^2$$

$$v_f = 17,26 \text{ m} \cdot \text{s}^{-1} \longrightarrow$$

**DOWNWARDS AS POSITIVE/
AFWAARTS AS POSITIEF**

$$v_f = v_i + a\Delta t \checkmark$$

$$17,26 = (0) + (9,8)\Delta t \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

OPTION 4/OPSIE 4

$$W_{\text{nc}} = \Delta K + \Delta U$$

$$W_{\text{nc}} = \Delta K + mg(h_f - h_i)$$

$$0 = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 + mgh_f - mgh_i$$

$$0 = \frac{1}{2}(v_f^2 - 0) + (9,8)(15,2)$$

$$v_f = 17,26 \text{ m} \cdot \text{s}^{-1} \longrightarrow$$

OR/OF

$$\Delta y = \left(\frac{v_i + v_f}{2} \right) \Delta t \checkmark$$

$$15,2 = \left(\frac{0 + 17,26}{2} \right) \Delta t \checkmark$$

$$\Delta t = 1,76 \text{ s } \checkmark$$

OPTION 5/OPSIE 5

$$W_{\text{net}} = \Delta E_k$$

$$w\Delta y \cos\theta = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$(9,8)(15,2)\cos 180^\circ = 0 - \frac{1}{2}(v_f)^2$$

$$v_f = 17,26 \text{ m} \cdot \text{s}^{-1} \longrightarrow$$

OPTION 6/OPSIE 6 UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF $F_{\text{net}}\Delta t = \Delta p = m(v_f - v_i)$ $mg\Delta t = m(v_f - v_i)$ $-9,8\Delta t = -17,26 - (0)$ ✓ $\Delta t = 1,76 \text{ s}$ ✓	DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF $F_{\text{net}}\Delta t = \Delta p = m(v_f - v_i)$ $mg\Delta t = m(v_f - v_i)$ $9,8\Delta t = 17,26 - (0)$ ✓ $\Delta t = 1,76 \text{ s}$ ✓
---	--

(3)

3.2.2

POSITIVE MARKING FROM QUESTION 3.2.1.**POSITIEWE NASIEN VANAF VRAAG 3.2.1.****Marking criteria/Nasienkriteria**

- Correct substitution to calculate Δt for ball A./Korrekte vervanging om Δt te bereken vir bal A. ✓
- Subtraction $1,76 - 0,81$./Aftrekking $1,76 - 0,81$. ✓
- Correct formula to calculate v_i for ball B./Korrekte formule om v_i te bereken vir bal B. ✓
- Correct substitution to calculate v_i for ball B./Korrekte vervanging om v_i te bereken vir bal B. ✓
- Final answer/Finale antwoord: $4,66 \text{ m}\cdot\text{s}^{-1}$ ✓ (4,66 to/tot 4,7)

OPTION 1/OPSIE 1**UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF**

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \checkmark$$

$$-3,2 = (0) + \frac{1}{2}(-9,8)(\Delta t)^2 \checkmark$$

$$\Delta t = 0,81 \text{ s}$$

$$\Delta t(B) = 1,76 - 0,81 \checkmark$$

$$\Delta t(B) = 0,95 \text{ s}$$

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$$

$$(0) = v_i(0,95) + \frac{1}{2}(-9,8)(0,95)^2 \checkmark$$

$$v_i = 4,66 \text{ m}\cdot\text{s}^{-1} \checkmark$$

DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \checkmark$$

$$3,2 = (0) + \frac{1}{2}(9,8)(\Delta t)^2 \checkmark$$

$$\Delta t = 0,81 \text{ s}$$

$$\Delta t(B) = 1,76 - 0,81 \checkmark$$

$$\Delta t(B) = 0,95 \text{ s}$$

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$$

$$0 = -v_i(0,95) + \frac{1}{2}(9,8)(0,95)^2 \checkmark$$

$$v_i = 4,66 \text{ m}\cdot\text{s}^{-1} \checkmark$$

OPTION 2/OPSIE 2**UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF**

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \checkmark$$

$$-3,2 = (0) + \frac{1}{2}(-9,8)(\Delta t)^2 \checkmark$$

$$\Delta t = 0,81 \text{ s}$$

$$\Delta t(B) = 1,76 - 0,81 \checkmark$$

$$\Delta t(B) = 0,95 \text{ s}$$

$$v_f = v_i + a\Delta t$$

$$-v_i = v_i + (-9,8)(0,95) \checkmark$$

$$v_i = 4,66 \text{ m}\cdot\text{s}^{-1} \checkmark$$

DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF

$$\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2 \checkmark$$

$$3,2 = (0) + \frac{1}{2}(9,8)(\Delta t)^2 \checkmark$$

$$\Delta t = 0,81 \text{ s}$$

$$\Delta t(B) = 1,76 - 0,81 \checkmark$$

$$\Delta t(B) = 0,95 \text{ s}$$

$$v_f = v_i + a\Delta t$$

$$v_i = -v_i + (9,8)(0,95) \checkmark$$

$$v_i = 4,66 \text{ m}\cdot\text{s}^{-1} \checkmark$$

Calculate/Bereken: $v_f = 7,92 \text{ m} \cdot \text{s}^{-1}$ OPTION 3/OPSIE 3 UPWARDS +/ OPWAARTS + $v_f^2 = v_i^2 + 2a\Delta y$ $v_f^2 = (0)^2 + (2)(-9,8)(-3,2)$ $v_f = -7,92 \text{ m} \cdot \text{s}^{-1}$ → DOWNWARDS +/ AFWAARTS + $v_f^2 = v_i^2 + 2a\Delta y$ $v_f^2 = (0)^2 + (2)(9,8)(3,2)$ $v_f = 7,92 \text{ m} \cdot \text{s}^{-1}$ →	Calculate/Bereken: $\Delta t(B) = 0,95 \text{ s}$ UPWARDS +/ OPWAARTS + $v_f = v_i + a\Delta t$ $-7,92 = (0) + (-9,8)\Delta t$ ✓ $\Delta t = 0,81 \text{ s}$ $\Delta t(B) = 1,76 - 0,81$ ✓ $\Delta t(B) = 0,95 \text{ s}$ → OR/OF $\Delta y = \left(\frac{v_i + v_f}{2}\right) \Delta t$ $-3,2 = \left(\frac{0 - 7,92}{2}\right) \Delta t$ ✓ $\Delta t = 0,81 \text{ s}$ $\Delta t(B) = 1,76 - 0,81$ ✓ $\Delta t(B) = 0,95 \text{ s}$ →	Calculate/Bereken: $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ UPWARDS +/ OPWAARTS + $(\Delta t_{\text{up and down}} = 0,95 \text{ s})$ $v_f = v_i + a\Delta t$ ✓ $-v_i = v_i + (-9,8)(0,95)$ ✓ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓ OR/OF $(\Delta t_{\text{up}} = 0,475 \text{ s})$ $v_f = v_i + a\Delta t$ ✓ $0 = v_i + (-9,8)(0,475)$ ✓ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓ OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓ $0 = v_i(0,95) + \frac{1}{2}(-9,8)(0,95)^2$ ✓ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓
OPTION 4/OPSIE 4 $(E_{\text{mech}})_{\text{top}} = (E_{\text{mech}})_{\text{bot}}$ $(E_p + E_k)_{\text{top}} = (E_p + E_k)_{\text{bot}}$ $(mgh + \frac{1}{2}mv_i^2)_{\text{top}} = (mgh + \frac{1}{2}mv_f^2)_{\text{bot}}$ $(9,8)(3,2) + 0 = 0 + (\frac{1}{2})(v_f)^2$ $v_f = 7,92 \text{ m} \cdot \text{s}^{-1}$ →	OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ $-12 = -7,92\Delta t + \frac{1}{2}(-9,8)\Delta t^2$ ✓ $\Delta t = 0,95 \text{ s}$ → DOWNWARDS +/ AFWAARTS + $v_f = v_i + a\Delta t$ $7,92 = (0) + (9,8)\Delta t$ ✓ $\Delta t = 0,81 \text{ s}$ $\Delta t(B) = 1,76 - 0,81$ ✓ $\Delta t(B) = 0,95 \text{ s}$ →	DOWNWARDS +/ AFWAARTS + $v_f = v_i + a\Delta t$ ✓ $v_i = v_i + (9,8)(0,95)$ ✓ $= -4,66$ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓
OPTION 5/OPSIE 5 $W_{\text{nc}} = \Delta K + \Delta U$ $W_{\text{nc}} = \Delta K + mg(h_f - h_i)$ $0 = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 + mgh_f - mgh_i$ $0 = \frac{1}{2}(v_f^2 - 0) + (9,8)(3,2)$ $v_f = 7,92 \text{ m} \cdot \text{s}^{-1}$ →	OR/OF $\Delta y = \left(\frac{v_i + v_f}{2}\right) \Delta t$ $3,2 = \left(\frac{0 + 7,92}{2}\right) \Delta t$ ✓ $\Delta t = 0,81 \text{ s}$ $\Delta t(B) = 1,76 - 0,81$ ✓ $\Delta t(B) = 0,95 \text{ s}$ → OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ $12 = 7,92\Delta t + \frac{1}{2}(9,8)\Delta t^2$ ✓ $\Delta t = 0,95 \text{ s}$ →	OR/OF $(\Delta t_{\text{up}} = 0,475 \text{ s})$ $v_f = v_i + a\Delta t$ ✓ $0 = v_i + (9,8)(0,475)$ ✓ $= -4,66$ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓ OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓ $0 = -v_i(0,95) + \frac{1}{2}(9,8)(0,95)^2$ ✓ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓
OPTION 6/OPSIE 6 $W_{\text{net}} = \Delta E_k$ $w\Delta y \cos\theta = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ $(9,8)(3,2)\cos 0^\circ = \frac{1}{2}v_f^2 - 0$ $v_f = 7,92 \text{ m} \cdot \text{s}^{-1}$ →	OR/OF $\Delta y = \left(\frac{v_i + v_f}{2}\right) \Delta t$ $3,2 = \left(\frac{0 + 7,92}{2}\right) \Delta t$ ✓ $\Delta t = 0,81 \text{ s}$ $\Delta t(B) = 1,76 - 0,81$ ✓ $\Delta t(B) = 0,95 \text{ s}$ → OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ $12 = 7,92\Delta t + \frac{1}{2}(9,8)\Delta t^2$ ✓ $\Delta t = 0,95 \text{ s}$ →	OR/OF $(\Delta t_{\text{up}} = 0,475 \text{ s})$ $v_f = v_i + a\Delta t$ ✓ $0 = v_i + (9,8)(0,475)$ ✓ $= -4,66$ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓ OR/OF $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$ ✓ $0 = -v_i(0,95) + \frac{1}{2}(9,8)(0,95)^2$ ✓ $v_i = 4,66 \text{ m} \cdot \text{s}^{-1}$ ✓

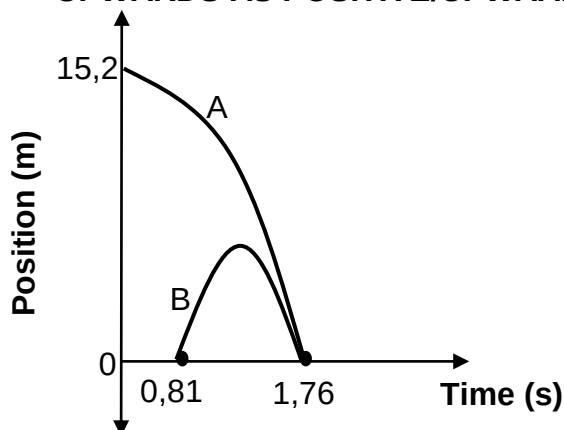
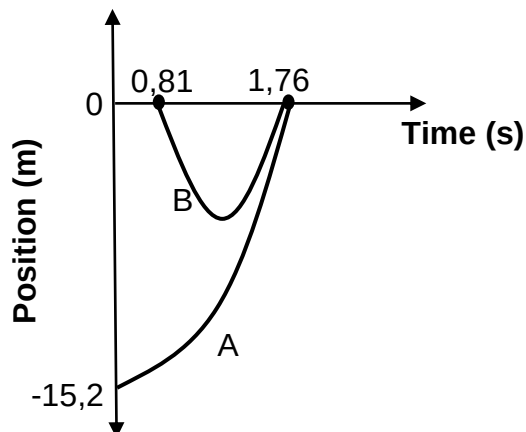
OPTION 7/OPSIE 7 UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF $F_{\text{net}}\Delta t = \Delta p$ $= m(v_f - v_i)$ $(-9,8)(0,95) \checkmark = -2v_i \checkmark \checkmark$ $v_i = 4,66 \text{ m}\cdot\text{s}^{-1} \checkmark (4,655 \text{ m}\cdot\text{s}^{-1})$	DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF $F_{\text{net}}\Delta t = \Delta p$ $= m(v_f - v_i)$ $(9,8)(0,95) \checkmark = 2v_i \checkmark \checkmark$ $v_i = 4,66 \text{ m}\cdot\text{s}^{-1} (4,655 \text{ m}\cdot\text{s}^{-1}) \checkmark$
--	---

(5)

3.3

POSITIVE MARKING FROM QUESTION 3.2.1/**POSITIEWE NASIEN VANAF VRAAG 3.2.1****Marking criteria/Nasienkriteria:**

- Initial position of ball A = 15,2 m and B = 0 m./Oorspronklike posisie van A = 15,2 m en B = 0 m. ✓
- Starting times for A = 0 s and B = 0,81 s./Begintye van A = 0 s en B = 0,81 s. ✓
- Both balls strike the ground at t = 1,76 s./Albei balle tref die grond op t = 1,76 s. ✓
- Shape of graph for ball A./Vorm van grafiek vir bal A. ✓
- Shape of graph for ball B./Vorm van grafiek vir bal B. ✓
- If graphs are drawn on separate axis/Indien grafieke op aparte asse geteken word:
max/maks: 4/5

UPWARDS AS POSITIVE/OPWAARTS AS POSITIEF (CHANGED GRAPH)**DOWNWARDS AS POSITIVE/AFWAARTS AS POSITIEF:**(5)
[15]

QUESTION 4 /VRAAG 4

4.1

Marking criteria/Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark/Indien enige van die onderstreepte sleutelwoorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

In an isolated system the total (linear) momentum is conserved/remains constant. ✓✓ (accept closed system)

In 'n geïsoleerde sisteem bly die totale (lineêre) momentum behoue/konstant. ✓✓ (aanvaar geslote sisteem)

ACCEPT FOR 1 MARK/AANVAAR VIR 1 PUNT

The total (linear) momentum before collision is equal to the total (linear) momentum after collision provided the system is isolated/the net external force on the system is zero.

Die totale (lineêre) momentum voor 'n botsing is gelyk aan die totale (lineêre) momentum na botsing mits die stelsel geïsoleer is/die netto eksterne krag op die stelsel is nul.

(2)

4.2.1

OPTION 1/OPSIE 1**RIGHT AS POSITIVE/REGS AS POSITIEF:**

$$\begin{aligned} \Sigma p_i &= \Sigma p_f \\ m_A v_{Ai} + m_B v_{Bi} &= m_A v_{Af} + m_B v_{Bf} \\ m_A v_{Ai} + m_B v_{Bi} &= (m_A + m_B) v_f \\ (7,2)(0,4) + (0) &= (7,2 + 5,3) v_f \quad \checkmark \\ v_f &= 0,23 \text{ m} \cdot \text{s}^{-1} \quad \checkmark \end{aligned} \quad \left. \vphantom{\begin{aligned} \Sigma p_i &= \Sigma p_f \\ m_A v_{Ai} + m_B v_{Bi} &= m_A v_{Af} + m_B v_{Bf} \\ m_A v_{Ai} + m_B v_{Bi} &= (m_A + m_B) v_f \\ (7,2)(0,4) + (0) &= (7,2 + 5,3) v_f \end{aligned}} \right\} \checkmark \text{ Any one/Enige een}$$

LEFT AS POSITIVE/LINKS AS POSITIEF:

$$\begin{aligned} \Sigma p_i &= \Sigma p_f \\ m_A v_{Ai} + m_B v_{Bi} &= m_A v_{Af} + m_B v_{Bf} \\ m_A v_{Ai} + m_B v_{Bi} &= (m_A + m_B) v_f \\ (7,2)(-0,4) + (0) &= (7,2 + 5,3) v_f \quad \checkmark \\ v_f &= -0,23 \\ \therefore v_f &= 0,23 \text{ m} \cdot \text{s}^{-1} \quad \checkmark \end{aligned} \quad \left. \vphantom{\begin{aligned} \Sigma p_i &= \Sigma p_f \\ m_A v_{Ai} + m_B v_{Bi} &= m_A v_{Af} + m_B v_{Bf} \\ m_A v_{Ai} + m_B v_{Bi} &= (m_A + m_B) v_f \\ (7,2)(-0,4) + (0) &= (7,2 + 5,3) v_f \end{aligned}} \right\} \checkmark \text{ Any one/Enige een}$$

OPTION 2/OPSIE 2**RIGHT AS POSITIVE/REGS AS POSITIEF**

$$\begin{aligned} \Delta p_{\text{trolley A}} &= -\Delta p_{\text{trolley B}} \\ m_A(v_{Af} - v_{Ai}) &= -m_B(v_{Bf} - v_{Bi}) \\ (7,2)(v_f - 0,4) &= -(5,3)(v_f - 0) \quad \checkmark \\ v_f &= 0,23 \text{ m} \cdot \text{s}^{-1} \quad \checkmark \end{aligned} \quad \left. \vphantom{\begin{aligned} \Delta p_{\text{trolley A}} &= -\Delta p_{\text{trolley B}} \\ m_A(v_{Af} - v_{Ai}) &= -m_B(v_{Bf} - v_{Bi}) \\ (7,2)(v_f - 0,4) &= -(5,3)(v_f - 0) \end{aligned}} \right\} \checkmark \text{ Any one/Enige een}$$

LEFT AS POSITIVE/LINKS AS POSITIEF

$$\begin{aligned} \Delta p_{\text{trolley A}} &= -\Delta p_{\text{trolley B}} \\ m_A(v_{Af} - v_{Ai}) &= -m_B(v_{Bf} - v_{Bi}) \\ (7,2)(v_f + 0,4) &= -(5,3)(v_f - 0) \quad \checkmark \\ v_f &= -0,23 \\ \therefore v_f &= 0,23 \text{ m} \cdot \text{s}^{-1} \quad \checkmark \end{aligned} \quad \left. \vphantom{\begin{aligned} \Delta p_{\text{trolley A}} &= -\Delta p_{\text{trolley B}} \\ m_A(v_{Af} - v_{Ai}) &= -m_B(v_{Bf} - v_{Bi}) \\ (7,2)(v_f + 0,4) &= -(5,3)(v_f - 0) \end{aligned}} \right\} \checkmark \text{ Any one/Enige een}$$

(3)

4.2.2

POSITIVE MARKING FROM QUESTIONS 4.2.1.**POSITIEWE NASIEN VANAF VRAAG 4.2.1.**

OPTION 1/OPSIE 1 RIGHT AS POSITIVE/ REGS AS POSITIEF: Force of B on A/Krag van B op A: $F_{\text{net}}\Delta t = \Delta p$ } ✓ Any one/ $F_{\text{net}}\Delta t = m(v_f - v_i)$ } Enige een $F_{\text{net}}(0,02) = 7,2(0,23 - 0,4)$ ✓ $F_{\text{net}} = -61,2$ $\therefore F_{\text{net}} = 61,2 \text{ N}$ ✓ (60,95 N to/tot 61,2 N)	LEFT AS POSITIVE/ LINKS AS NEGATIEF: Force of B on A/Krag van B op A: $F_{\text{net}}\Delta t = \Delta p$ } ✓ Any one/ $F_{\text{net}}\Delta t = m(v_f - v_i)$ } Enige een $F_{\text{net}}(0,02) = 7,2(-0,23 + 0,4)$ ✓ $F_{\text{net}} = 61,2 \text{ N}$ ✓ (60,95 N to/tot 61,2 N)
OPTION 2/OPSIE 2 RIGHT AS POSITIVE/ REGS AS POSITIEF: Force of A on B/Krag van A op B: $F_{\text{net}}\Delta t = \Delta p$ } ✓ Any one/ $F_{\text{net}}\Delta t = m(v_f - v_i)$ } Enige een $F_{\text{net}}(0,02) = 5,3(0,23 - 0)$ ✓ $F_{\text{net}} = 60,95 \text{ N}$ ✓ (60,95 N to/tot 61,2 N)	LEFT AS POSITIVE/ LINKS AS POSITIEF: Force of A on B/Krag van A op B: $F_{\text{net}}\Delta t = \Delta p$ } ✓ Any one/ $F_{\text{net}}\Delta t = m(v_f - v_i)$ } Enige een $F_{\text{net}}(0,02) = 5,3(-0,23 - 0)$ ✓ $F_{\text{net}} = -60,95$ $\therefore F_{\text{net}} = 60,95 \text{ N}$ ✓ (60,95 N to/tot 61,2 N)
OPTION 3/OPSIE 3 RIGHT AS POSITIVE/REGS AS POSITIEF A: $v_f = v_i + a\Delta t$ $0,23 = 0,4 + a(0,02)$ $a = -8,5 \text{ m}\cdot\text{s}^{-2}$ $F_{\text{net}} = ma$ ✓ $= (7,2)(-8,5)$ ✓ $= -61,20$ $F_{\text{net}} = 61,20 \text{ N}$ ✓	LEFT AS POSITIVE/LINKS AS POSITIEF A: $v_f = v_i + a\Delta t$ $-0,23 = -0,4 + a(0,02)$ $a = 8,5 \text{ m}\cdot\text{s}^{-2}$ $F_{\text{net}} = ma$ ✓ $= (7,2)(8,5)$ ✓ $= 61,20 \text{ N}$ $F_{\text{net}} = 61,20 \text{ N}$ ✓
OPTION 4/OPSIE 4 RIGHT AS POSITIVE/REGS AS POSITIEF B: $v_f = v_i + a\Delta t$ $0,23 = 0 + a(0,02)$ $a = 11,5 \text{ m}\cdot\text{s}^{-2}$ $F_{\text{net}} = ma$ ✓ $= (5,3)(11,5)$ ✓ $= 60,95 \text{ N}$ $F_{\text{net}} = 60,95 \text{ N}$ ✓	LEFT AS POSITIVE/LINKS AS POSITIEF B: $v_f = v_i + a\Delta t$ $-0,23 = 0 + a(0,02)$ $a = -11,5 \text{ m}\cdot\text{s}^{-2}$ $F_{\text{net}} = ma$ ✓ $= (5,3)(-11,5)$ ✓ $= -60,95 \text{ N}$ $F_{\text{net}} = 60,95 \text{ N}$ ✓

(3)
[8]

QUESTION 5/VRAAG 5

5.1

Marking criteria/Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark./Indien enige van die onderstreepte sleutelwoorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

A force is non-conservative if the work it does/done on an object which is moving between two points depends on the path taken. ✓✓

'n Krag is nie-konserwatief indien die arbeid wat dit verrig/doen op 'n voorwerp wat tussen twee punte beweeg afhanklik is van die pad gevolg.

OR/OF

A force is non-conservative if the work it does/done in moving an object around a closed path is non-zero. ✓✓

'n Krag is nie-konserwatief indien die arbeid wat dit verrig/doen om 'n voorwerp op 'n geslote pad te beweeg, nie nul is nie.

(2)

Note/Aantekening:

If the word 'work' is omitted, 0/2.

Indien die woord 'arbeid' weggelaat is, 0/2.

5.2

Marking criteria/Nasienkriteria

- Any one of the correct equations./Enige een van die korrekte vergelykings. ✓
- Correct substitution for work done by gravity or ΔU . ✓
Korrekte vervanging vir arbeid verrig deur gravitasie of ΔU .
- Correct substitution for work done by motor and friction./Korrekte vervanging aan van arbeid verrig deur motor en wrywing. ✓
- Correct substitution for ΔK ./Korrekte vervanging vir ΔK . ✓
- Correct final answer./Korrekte finale antwoord: $5,96 \text{ m}\cdot\text{s}^{-1}$ ✓

OPTION 1/OPSIE 1

$$\left. \begin{aligned} W_{\text{net}} &= \Delta K \\ W_w + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ mgsin\theta\Delta x\cos\theta + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$$

$$(20)(9,8)(\sin 18^\circ)(15,6)\cos 180^\circ + (13,5)(15,6)\cos 180^\circ + (96,8)(15,6)\cos 0^\circ = \frac{1}{2}(20)(v_f^2 - 0^2) \checkmark$$

$$v_f = 5,96 \text{ m}\cdot\text{s}^{-1} \checkmark$$

OPTION 2/OPSIE 2

$$\left. \begin{aligned} W_{\text{nc}} &= \Delta K + \Delta U \\ W_f + W_F &= \Delta K + mg(h_f - h_i) \\ f\Delta x\cos\theta + F\Delta x\cos\theta &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 + mgh_f - mgh_i \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$$

$$(13,5)(15,6)\cos 180^\circ + 96,8(15,6)\cos 0^\circ = \frac{1}{2}(20)(v_f^2 - 0^2) + (20)(9,8)(15,6 \sin 18^\circ - 0) \checkmark$$

$$v_f = 5,96 \text{ m}\cdot\text{s}^{-1} \checkmark$$

OPTION 3/OPSIE 3

$$\left. \begin{aligned} W_{\text{net}} &= \Delta K \\ W_w + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ -\Delta E_p + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$$

$$-(20)(9,8)(15,6 \sin 18^\circ) + (13,5)(15,6)\cos 180^\circ + (96,8)(15,6)\cos 0^\circ = \frac{1}{2}(20)(v_f^2 - 0^2) \checkmark$$

$$v_f = 5,96 \text{ m}\cdot\text{s}^{-1} \checkmark$$

OPTION 4/OPSIE 4

$$\begin{aligned}
 W_{\text{net}} &= \Delta K \\
 W_w + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark \text{ Any one/Enige} \\
 mg\Delta x \cos\theta + W_f + W_F &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\
 (20)(9,8)(15,6)\cos 108^\circ + (13,5)(15,6)\cos 180^\circ + (96,8)(15,6)\cos 0^\circ &= \frac{1}{2}(20)(v_f^2 - 0^2) \checkmark \\
 v_f &= 5,96 \text{ m}\cdot\text{s}^{-1} \checkmark
 \end{aligned}$$

OPTION 5/OPSIE 5

$$\begin{aligned}
 F_{\text{net}} &= ma \\
 F_{\text{net}} &= F - F_{w//} - f \\
 &= 96,8 - (20)(9,8)\sin 18^\circ \checkmark - 13,5 \\
 &= 22,73 \text{ N} \\
 W_{\text{net}} &= \Delta K \\
 F_{\text{net}}\Delta x \cos\theta &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark \text{ Any one/Enige} \\
 22,73(15,6)\cos 0^\circ &= \frac{1}{2}(20)(v_f^2 - 0^2) \checkmark \\
 v_f &= 5,96 \text{ m}\cdot\text{s}^{-1} \checkmark
 \end{aligned}$$

(5)

5.3

POSITIVE MARKING FROM QUESTION 5.2.**POSITIEWE NASIEN VANAF VRAAG 5.2.****Marking criteria/Nasienkriteria**

- Correct equation for power/Korrekte vergelyking vir drywing. $\checkmark$
- Correct substitution into power equation./ $\checkmark$
Korrekte vervanging in drywingvergelgking.
- Correct final answer/Korrekte finale antwoord: 288,46 W $\checkmark$
Range: 286,46 W to/na 288,73 W

OPTION 1/OPSIE 1

$$\begin{aligned}
 P_{\text{ave}} &= Fv_{\text{ave}} \checkmark \\
 &= 96,8 \left(\frac{(0) + 5,96}{2} \right) \checkmark \\
 &= 288,46 \text{ W} \checkmark
 \end{aligned}$$

OPTION 2/OPSIE 2

$$\begin{aligned}
 \Delta x &= \left(\frac{v_i + v_f}{2} \right) \Delta t \\
 15,6 &= \left(\frac{(0) + 5,96}{2} \right) \Delta t \\
 \Delta t &= 5,23 \text{ s } (5,24) \longrightarrow \\
 P &= \frac{W}{\Delta t} \\
 P &= \frac{F\Delta x \cos \Theta}{\Delta t} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark \text{ Any one/Enige een} \\
 P &= \frac{96,8(15,6) \cos 0^\circ}{5,23} \checkmark \\
 P &= 288,73 \text{ W} \checkmark
 \end{aligned}$$

OPTION 3/OPSIE 3

$$\begin{aligned}
 v_f^2 &= v_i^2 + 2a\Delta x \\
 (5,96)^2 &= 0^2 + 2a(15,6) \\
 a &= 1,14 \text{ m}\cdot\text{s}^{-2} (1,13) \\
 v_f &= v_i + a\Delta t \\
 5,96 &= 0 + (1,14)\Delta t \\
 \Delta t &= 5,23 \text{ s } (5,27) \longrightarrow
 \end{aligned}$$

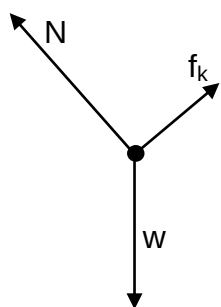
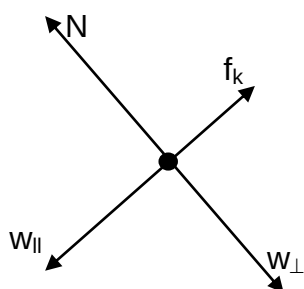
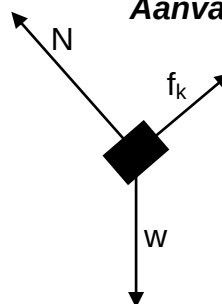
OR/OF

$$\begin{aligned}
 v_{\text{ave}} &= \frac{\Delta x}{\Delta t} \\
 &= \frac{15,6 - 0}{5,23 - 0} \\
 &= 2,98 \text{ m}\cdot\text{s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 P_{\text{ave}} &= Fv_{\text{ave}} \checkmark \\
 &= 96,8(2,98) \checkmark \\
 &= 288,46 \text{ W} \checkmark
 \end{aligned}$$

(3)

5.4

**Accept/Aanvaar:****Accept force diagram/
Aanvaar kragte-diagram:**

	Accepted symbols/Aanvaarde simbole
N ✓	$F_N/196\text{N}/\text{Normal}/\text{Normaal}/F_{\text{normal}}/F_{\text{normaal}}$
f ✓	(kinetic) friction/(kineties) wrywing/ $F_f/F_w/f_k/f_r$
w ✓	$F_g/F_w/\text{weight}/\text{gewig}/mg/\text{gravitational force}/\text{gravitasiekrag}/F_{\text{Earth on crate}}/F_{\text{Aarde op krat}}$
Notes/Aantekeninge: - Accept correct numerical values for the forces./Aanvaar korrekte numeriese waardes vir die kragte. - Mark awarded for label <u>and</u> arrow./Punt toegeken vir benoeming <u>en</u> pyltjie. - Do not penalise for length of arrows since drawing is not to scale./Moenie vir die lengte van die pyltjies penaliseer nie aangesien die tekening nie volgens skaal is nie. - Any other additional force(s)/Enige ander addisionele krag(te): Max/Maks $2/3$ - If everything correct, but no arrows/Indien alles korrek, maar geen pyltjies: Max/Maks $2/3$	

(3)

5.5

POSITIVE MARKING FROM QUESTION 5.2.**POSITIEWE NASIEN VANAF VRAAG 5.2.****Marking criteria/Nasienkriteria:**

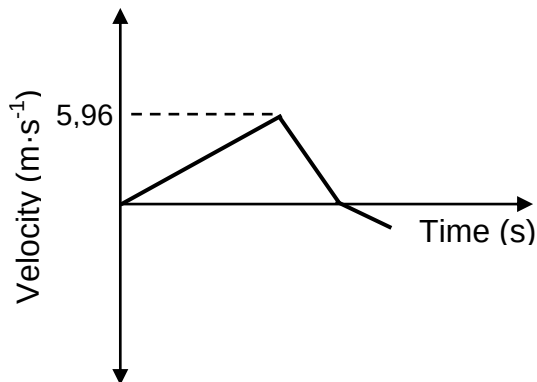
- First straight line starting at zero with positive gradient, reaching a maximum velocity./Eerste reguit lyn met 'n positiewe gradiënt begin by nul en bereik maksimum snelheid. ✓
- Second straight line with negative gradient from maximum velocity to zero./Tweede reguit lyn met negatiewe gradiënt vanaf maksimum snelheid na zero. ✓
- Third straight line continuing from second line at zero and extending below the x-axis./Derde reguit lyn wat aangaan vanaf tweede lyn vanaf nul en verleng onder die x-as. ✓
- Third line has a smaller negative gradient than the second line./Derde reguit lyn het 'n kleiner negatiewe gradiënt as die tweede lyn ✓

Note/Aantekening:

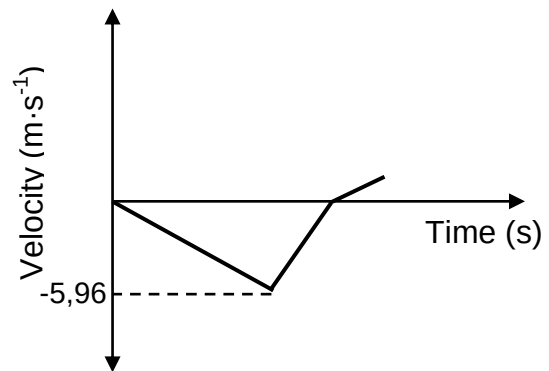
Direction of gradients opposite for graph 2./Rigting van hellings teenoorgesteld vir grafiek 2.

No marks given for values of velocities./Geen punte toegeken vir waardes van snelhede nie.

UPWARDS AS POSITIVE
OPWAARTS AS POSITIEF:



DOWNWARDS AS POSITIVE
AFWAARTS AS POSITIEF:



(4)
[17]

QUESTION 6/VRAAG 6

6.1.1

Marking criteria/Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark./Indien enige van die onderstreepte sleutel woorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The (apparent) change in frequency (or pitch) (of the sound) detected by a listener because the source and the listener have different velocities relative to the medium of propagation. ✓✓

Die (skynbare) verandering in die frekwensie (of toonhoogte) (van die klank) waargeneem deur 'n luisteraar omdat die bron en die luisteraar verskillende snelhede relatief tot die voortplantingsmedium het.

OR/OF

An (apparent) change in observed/detected frequency/pitch as a result of the relative motion between a source and an observer/listener.

'n (Skynbare) verandering in waargenome frekwensie/toonhoogte as gevolg van die relatiewe beweging tussen die bron en 'n waarnemer/luisteraar.

(2)

6.1.2

$$f_L = \frac{v \pm v_L}{v \pm v_s} f_s \quad \checkmark \quad \text{OR/OF} \quad f_L = \frac{v + v_L}{v} f_s$$

$$f_L = \frac{340 + 22}{340} \times 24\,000 \quad \checkmark$$

$$f_L = 25\,552,94 \text{ Hz}$$

$$f_L = \frac{v \pm v_L}{v \pm v_s} f_s \quad \text{OR/OF} \quad f_L = \frac{v}{v - v_s} f_s$$

$$f_L = \frac{340}{340 - 22} \times 25\,552,94 \quad \checkmark$$

$$f_L = 27\,320,75 \text{ Hz} \quad \checkmark$$

(6)

6.2

The frequencies of the spectral lines would have decreased. / Die frekwensies van die spektrale lyne sou verminder het. ✓✓

OR/OF

The spectral lines from the distant star are shifted towards lower frequency end of the spectrum. / Die spektrale lyne van die ver af ster sou verskuif na 'n laer frekwensie op die spektrum.

(2)
[10]**QUESTION 7 / VRAAG 7**

7.1

Marking criteria / Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark. / Indien enige van die onderstreepte sleutelwoorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

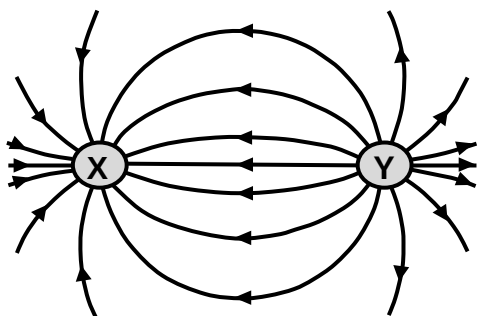
If any reference is made to mass / Indien enige verwysing gemaak is na massa: 0/2

The magnitude of the electrostatic force exerted by one (stationary) point charge (Q_1) on another (stationary) point charge (Q_2) is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance (r) between them. ✓✓

Die grootte van die elektrostatiese krag uitgeoefen deur een (stilstaande) puntlading (Q_1) op 'n ander puntlading (Q_2) is direk eweredig aan die produk van die grootte van die ladings en omgekeerd eweredig aan die kwadraat van die afstand (r) tussen hulle.

(2)

7.2

**Criteria for graph/Kriteria vir grafiek:**

Correct shape/ Korrekte vorm	✓
Correct direction from Y to X. /Korrekte rigting van Y na X.	✓
Lines must not cross and must touch charges./ Lyne mag nie kruis nie en moet die ladings raak.	✓

Note/Aantekening:

- If the net electric field pattern is drawn for two like charges: $0/3$
Indien die netto elektriese veldpatroon vir twee gelyksoortige ladings geteken is: $0/3$
- Ignore labels for point charges/ Ignoreer byskrifte vir puntladings.

(3)

7.3

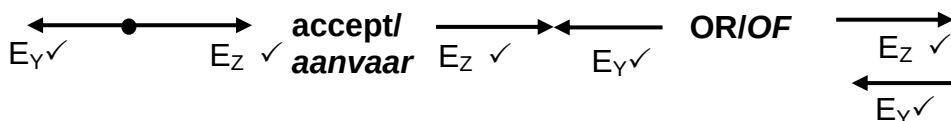
$$F = \frac{kQ_Y Q_X}{r^2} \checkmark$$

$$F = \frac{(9 \times 10^9)(7,2 \times 10^{-9})(7,2 \times 10^{-9})}{(0,03)^2} \checkmark$$

$$= 5,18 \times 10^{-4} \text{ N} \checkmark \quad (0,000518 \text{ N})$$

(3)

7.4

**Notes/Aantekeninge:**

- 1 Mark for arrows in opposite directions./1 Punt vir pyle in teenoorgestelde rigtings.
- 1 Mark for correct labels./1 Punt vir korrekte benoemings.
- Do not penalise for length of arrow since drawing is not to scale./Moenie penaliseer vir die lengte van die pyltjie nie aangesien tekening nie volgens skaal is nie.
- Accept Y and Z as labels./Aanvaar Y en Z as benoeming.

(2)

7.5

OPTION 1/OPSIE 1**Marking criteria/Nasienkriteria**

- Formula for electric field./Formule vir elektriese veld. ✓
- Substitution of E_{net} ./Vervanging van E_{net} . ✓
- Correct substitution into $\frac{kQ}{r^2}$ equation for charge Z or charge Y ✓
Korrekte vervanging in $\frac{kQ}{r^2}$. vergelyking vir lading Z of lading Y
- Subtraction of electric fields ($E_z + E_y$)/Aftrek van elektriese velde ($E_z + E_y$) ✓
- Correct final answer ✓/Korrekte finale antwoord: $6,25 \times 10^{-9} \text{C}$
Range/Gebied: $6,25 \times 10^{-9} \text{C}$ to/na $6,26 \times 10^{-9} \text{C}$

OPTION 1/OPSIE 1

$$E_{\text{net}} = E_z + E_y$$

$$= E_z - E_y$$

$$E_{\text{net}} = \frac{kQ_z}{r^2} + \frac{kQ_y}{r^2}$$

$$4,91 \times 10^5 \checkmark = \left(\frac{(9 \times 10^9)(Q_z)}{0,01^2} \right) \checkmark - \left(\frac{(9 \times 10^9)(7,2 \times 10^{-9})}{0,03^2} \right) \checkmark$$

$$Q_z = 6,25 \times 10^{-9} \text{C} \checkmark$$

OPTION 2/OPSIE 2**POSITIVE MARKING FROM QUESTION 7.3/****POSITIEWE NASIEN VANAF VRAAG 7.3****Marking criteria/Nasienkriteria**

- Correct formula for electric field./Korrekte formule vir elektriese veld. ✓
- Substitution of F_{net} ./Vervanging van F_{net} . ✓
- Correct substitution into $\left(\frac{kQ_z Q_x}{r^2} \right)$ for charge Z. ✓
Korrekte vervanging in $\left(\frac{kQ_z Q_x}{r^2} \right)$ vir lading Z
- Subtraction of forces ✓./Aftrek van kragte.
- Correct final answer ✓/Korrekte finale antwoord: $6,26 \times 10^{-9} \text{C}$ ✓
Range/Gebied: $6,25 \times 10^{-9} \text{C}$ to/na $6,26 \times 10^{-9} \text{C}$

$$E = \frac{F}{Q} \checkmark$$

$$4,91 \times 10^5 = \frac{F}{7,2 \times 10^{-9}}$$

$$F_{\text{net}} = 3,54 \times 10^{-3} \text{N}$$

$$F_{\text{net}} = F_{Z \text{ on } X} + F_{Y \text{ on } X}$$

$$F_{\text{net}} = \left(\frac{kQ_z Q_x}{r^2} \right) + \left(\frac{kQ_y Q_x}{r^2} \right)$$

$$(3,54 \times 10^{-3}) \checkmark = \left(\frac{(9 \times 10^9)(7,2 \times 10^{-9})Q_z}{(0,01)^2} \right) \checkmark - (5,18 \times 10^{-4})$$

$$Q_z = 6,26 \times 10^{-9} \text{C} \checkmark$$

(5)
[15]

QUESTION 8/VRAAG 8

8.1

Marking criteria/Nasienkriteria

If any of the underlined key words/phrases in the **correct context** is omitted deduct 1 mark./Indien enige van die onderstreepte sleutelwoorde/frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The potential difference across a conductor is directly proportional to the current in the conductor at constant temperature (provided all other physical conditions remain constant). ✓✓

Die potensiaalverskil oor 'n geleier is direk eweredig aan die stroom in die geleier by konstante temperatuur (mits alle ander fisiese toestande konstant bly).

OR/OF

The ratio of potential difference to current is constant at constant temperature.

Die verhouding van potensiaalverskil tot stroom is konstant by konstante temperatuur.

OR/OF

The current in a conductor is directly proportional to the potential difference across the conductor at constant temperature (provided all other physical conditions remain constant).

Die stroom in 'n geleier is direk eweredig aan die potensiaalverskil oor 'n geleier by konstante temperatuur (mits alle fisiese toestande konstant bly).

(2)

8.2.1

OPTION 1/OPSIE 1

$$R = \frac{V}{I} \checkmark$$

$$7 = \frac{V}{1,5} \checkmark$$

$$V = 10,5 \text{ V} \checkmark$$

OPTION 2/OPSIE 2

$$R = \frac{V}{I} \checkmark$$

$$2 = \frac{V}{1,5}$$

$$V = 3 \text{ V}$$

$$5 = \frac{V}{1,5}$$

$$V = 7,5 \text{ V}$$

$$V_T = V_1 + V_2$$

$$= 3 + 7,5 \quad (\text{addition of calculated values/optel van berekende waardes})$$

$$= 10,5 \text{ V} \checkmark$$

OPTION 3/OPSIE 3

Ratio of R:

$$3 : 7$$

Ratio of I:

$$7 : 3$$

$$I = 1,5 \times \frac{7}{3} = 3,5 \text{ A}$$

$$V = IR \checkmark$$

$$= (3,5)(3) \checkmark$$

$$= 10,5 \text{ V} \checkmark$$

(3)

8.2.2

POSITIVE MARKING FROM QUESTION 8.2.1.**POSITIEWE NASIEN VANAF VRAAG 8.2.1.****OPTION 1/OPSIE 1**

$$R = \frac{V}{I_3} \checkmark$$

$$3 = \frac{10,5}{I_3} \checkmark$$

$$I_3 = 3,5 \text{ A}$$

$$I_T = 1,5 + 3,5 \quad (\text{addition of calculated values/optel van berekende waardes})$$

$$= 5 \text{ A} \checkmark$$

OPTION 2/OPSIE 2

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

OR/OF

$$\frac{1}{R_p} = \frac{1}{7} + \frac{1}{3} \checkmark$$

$$R_p = 2,1 \Omega$$

$$R = \frac{V}{I_T} \checkmark$$

$$2,1 = \frac{10,5}{I_T} \checkmark$$

$$I_T = 5 \text{ A} \checkmark$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_p = \frac{(7)(3)}{7 + 3} \checkmark$$

$$= 2,1 \Omega$$

OPTION 3/OPSIE 3

$$I_{3\Omega} = \frac{7}{3} \times 1,5 \checkmark$$

$$= 3,5 \text{ A} \checkmark$$

$$I_{\text{total}} = 3,5 + 1,5$$

$$= 5 \text{ A} \checkmark$$

OPTION 4/OPSIE 4

$$I_s = \left(\frac{R_{3\Omega}}{R_{3\Omega} + R_S} \right) \times I_{\text{total}} \checkmark$$

$$1,5 \checkmark = \frac{3}{3+7} \checkmark \times I_{\text{total}}$$

$$I_{\text{total}} = 5 \text{ A} \checkmark$$

(4)

8.2.3 **POSITIVE MARKING FROM QUESTION 8.2.1 AND/OR QUESTION 8.2.2.**
POSITIEWE NASIEN VANAF VRAAG 8.2.1 EN/OF VRAAG 8.2.2.

<u>OPTION 1/OPSIE 1</u> $P = \frac{V^2}{R} \checkmark$ $= \frac{10,5^2}{3} \checkmark$ $= 36,75 \text{ W} \checkmark$	<u>OPTION 2/OPSIE 2</u> $P = I^2 R \checkmark$ $= (3,5)^2 (3) \checkmark$ $= 36,75 \text{ W} \checkmark$
<u>OPTION 3/OPSIE 3</u> $P = VI \checkmark$ $= (10,5)(3,5) \checkmark$ $= 36,75 \text{ W} \checkmark$	

(3)

8.3 **POSITIVE MARKING IF OPTION 2 IS USED IN QUESTION 8.2.2./**
POSITIEWE NASIEN INDIEN OPSIE 2 GEBRUIK IS IN VRAAG 8.2.2.

<u>Marking criteria/Nasienkriteria</u> - Correct equation for emf./Korrekte vergelyking vir emk. ✓ - Correct substitution for S_2 closed./Korrekte vervanging vir S_2 gesluit. ✓ - Correct substitution for S_2 open./Korrekte vervanging vir S_2 oop. ✓ - Equating emf or internal resistance equations. ✓ <i>Gelykstelling van emk of interne weerstand vergelykings.</i> - Correct final answer/Korrekte finale antwoord: 12,05 V ✓ Range/Gebied: 12,04 V to/tot 12,05 V	
$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_p} = \frac{1}{7} + \frac{1}{3}$ $R_p = 2,1 \Omega$ For S_1 and S_2 closed: $\begin{aligned} \mathcal{E} &= I(R + r) \\ \mathcal{E} &= IR + Ir \\ \mathcal{E} &= V_{\text{int}} + V_{\text{ext}} \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark \text{ Any one/ Enige een}$ $= 5(2,1 + r) \checkmark \dots (1)$	
For S_2 open: $\begin{aligned} \mathcal{E} &= I(R + r) \\ &= 3,64(3 + r) \checkmark \dots (2) \end{aligned}$	
(1) = (2) $5(2,1 + r) = 3,64(3 + r)$ $r = 0,31 \Omega$ $\mathcal{E} = 5(2,1 + 0,31)$ $= 12,05 \text{ V} \checkmark \quad (12,04)$ OR/OF $\mathcal{E} = 3,64(3 + 0,31)$ $= 12,05 \text{ V} \checkmark \quad (12,04)$	(1) = (2): $\frac{\mathcal{E} - 10,5}{5} = \frac{\mathcal{E} - 10,92}{3,64}$ $\mathcal{E} = 12,05 \text{ V} \checkmark \quad (12,04)$

(5)

8.4 Increases/Toeneem ✓

(Total) resistance increases./ (Totale) weerstand neem toe. ✓

Current decreases./ Stroom neem af. ✓

 V_{internal} / Internal volts decreases. / V_{intern} / Interne volts neem af. ✓

(4)

Note/Aantekening:

Award marks if learner proved the statement using calculated numerical values./

Ken punte toe indien leerder die stelling bewys deur berekende waardes te gebruik.

[21]

QUESTION 9/VRAAG 9

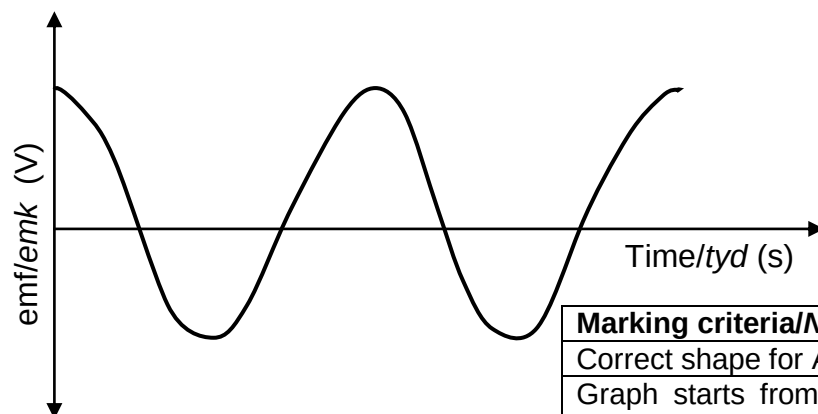
9.1.1 North pole/Noord-pool ✓

(1)

9.1.2 Y to X/Y na X ✓

(1)

9.1.3

**Marking criteria/Nasienkriteria**

Correct shape for AC./Korrekte vorm vir WS. ✓

Graph starts from maximum value./Grafiek begin by maksimum waarde. ✓

Two complete waves/Twee volledige golwe ✓

Note/Aantekening:

Accept graph starting at negative max./Aanvaar grafiek wat by negatiewe maks begin.

(3)

9.2.1

Marking criteria/Nasienkriteria:

- Formula to calculate V_{max} or I_{rms} ./Formule om V_{maks} of I_{wgk} te bereken. ✓
- Correct substitution of V_{rms} or I_{max} ./Korrekte vervanging van V_{wgk} of I_{maks} . ✓
- Correct substitution to calculate R ./Korrekte vervanging om R te bereken. ✓
- Correct final answer/Korrekte finale antwoord: 47,14 Ω to/tot 47,2 Ω ✓

OPTION 1/OPSIE 1

$$V_{\text{rms}} = \frac{V_{\text{max}}}{\sqrt{2}} \quad \checkmark$$

$$200 = \frac{V_{\text{max}}}{\sqrt{2}} \quad \checkmark$$

$$V_{\text{max}} = 282,84 \text{ V}$$

$$R = \frac{V}{I} \quad \downarrow$$

$$= \frac{282,84}{6} \quad \checkmark$$

$$R = 47,14 \Omega \quad \checkmark$$

OPTION 2/OPSIE 2

$$I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}} \quad \checkmark$$

$$I_{\text{rms}} = \frac{6}{\sqrt{2}} \quad \checkmark$$

$$I_{\text{rms}} = 4,24 \text{ A}$$

$$R = \frac{V}{I} \quad \downarrow$$

$$= \frac{200}{4,24} \quad \checkmark$$

$$R = 47,17 \Omega \quad \checkmark$$

OPTION 3/OPSIE 3 $I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}} \checkmark$ $I_{\text{rms}} = \frac{6}{\sqrt{2}} \checkmark$ $I_{\text{rms}} = 4,24 \text{ A}$ $P_{\text{ave}} = V_{\text{rms}} I_{\text{rms}}$ $= (200)(4,24)$ $= 848 \text{ W } (848,53)$	$P_{\text{ave}} = \frac{V_{\text{rms}}^2}{R}$ $848 = \frac{(200)^2}{R} \checkmark$ $R = 47,17 \Omega \checkmark$ OR/OF $P_{\text{ave}} = I_{\text{rms}}^2 R$ $848 = (4,24)^2 R \checkmark$ $R = 47,17 \Omega \checkmark$
--	--

(4)

9.2.2 POSITIVE MARKING FROM QUESTION 9.2.1.

POSITIEWE NASIEN VANAF VRAAG 9.2.1.

OPTION 1/OPSIE 1 $W = I^2 R \Delta t \checkmark$ $= (4,24)^2 (47,17) \checkmark (7200) \checkmark$ $= 6,11 \times 10^6 \text{ J } \checkmark (6,10 \times 10^6)$	OPTION 2/OPSIE 2 $W = VI \Delta t \checkmark$ $= (200)(4,24) \checkmark (7200) \checkmark$ $= 6,11 \times 10^6 \text{ J } \checkmark$
OPTION 3/OPSIE 3 $W = \frac{V^2 \Delta t}{R} \checkmark$ $= \frac{(200)^2}{47,27} (7200) \checkmark$ $= 6,11 \times 10^6 \text{ J } \checkmark (6,10 \times 10^6)$	OPTION 4/OPSIE 4 $P = \frac{W}{\Delta t} \checkmark$ $848 = \frac{W}{7200} \checkmark$ $= 6,11 \times 10^6 \text{ J } \checkmark$

(4)

[13]**QUESTION 10/VRAAG 10**

10.1 $6,63 \times 10^{-34} \checkmark$

(1)

10.2 **Marking criteria/Nasienkriteria**

If any of the underlined key words/phrases in the **correct context** is omitted or extra incorrect words added, deduct 1 mark./Indien enige van die onderstreepte sleutel woorde/frases in die **korrekte konteks** uitgelaat of onkorrekte woorde bygevoeg is, trek 1 punt af.

The minimum energy needed to eject an electron from a (metal) surface. $\checkmark \checkmark$
 Die minimum energie benodig om 'n elektron uit 'n (metaal)oppervlak vry te stel.

(2)

10.3.1 $W_0 = hf_0 \checkmark$
 $= (6,63 \times 10^{-34})(5 \times 10^{14}) \checkmark$
 $= 3,32 \times 10^{-19} \text{ J } \checkmark$

(3)

10.3.2 **POSITIVE MARKING FROM QUESTION 10.3.1./ POSITIEWE NASIEN VANAF VRAAG 10.3.1.**

OPTION 1/OPSIE 1

$$\text{Gradient} = \frac{\Delta E_k}{\Delta f} \checkmark$$

$$6,63 \times 10^{-34} \checkmark = \frac{(X - 0)}{(12,54 \times 10^{14} - 5 \times 10^{14})} \checkmark$$

$$X = 5 \times 10^{-19} \text{ (J)} \checkmark$$

OPTION 2/OPSIE 2

$$\left. \begin{aligned} E &= W_0 + E_{k(\max)} \\ hf &= hf_0 + \frac{1}{2}mv_{\max}^2 \\ h\frac{c}{\lambda} &= h\frac{c}{\lambda_0} + E_{k(\max)} \end{aligned} \right\} \checkmark \text{ Any one/Enige een}$$

$$\frac{(6,63 \times 10^{-34})(12,54 \times 10^{14})}{E_{k(\max)}} \checkmark = \frac{3,32 \times 10^{-19} + E_{k(\max)}}{E_{k(\max)}} \checkmark$$

$$E_{k(\max)} = 5 \times 10^{-19} \text{ J} \checkmark$$

$$X = 5 \times 10^{-19} \text{ (J)}$$

Note/Aantekening:

Do not penalise learner again in 10.3.2 if $(\times 10^{14})$ is omitted in 10.3.1/Moenie leerder weer penaliseer in 10.3.2 indien $(\times 10^{14})$ uitgelaat is in 10.3.1 nie.

(4)

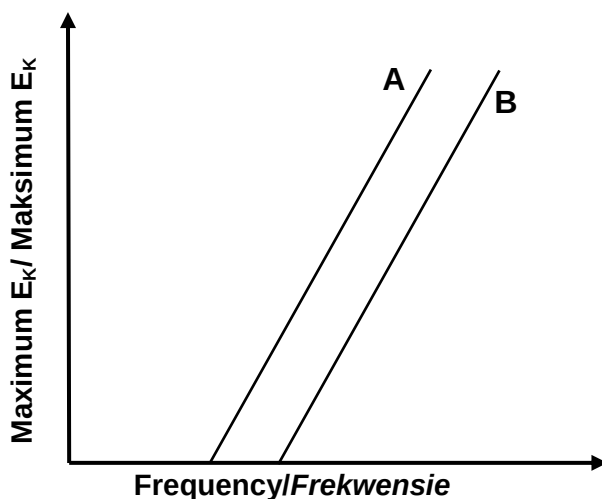
10.4.1 No effect/Geen effek $\checkmark$

(1)

10.4.2 Increases/Toeneem $\checkmark$

(1)

10.5



Marking criteria/Nasienkriteria	
Graph B to the right of graph A./Grafiek B aan regterkant van grafiek A.	$\checkmark$
Lines are parallel./Lyne is parallel.	$\checkmark$
If both graphs are not labelled/Indien beide grafieke nie benoem is nie: $0/2$	
If two separate graphs are drawn/Indien twee aparte grafieke geteken is: $0/2$	

(2)

[14]

TOTAL/TOTAAL: 150